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Music listening as a means of stress reduction in daily life

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Summary The relation between music listening and stress is inconsistently reported across studies, with the major part of studies being set in experimental settings. Furthermore, the psychobiological mechanisms for a potential stress-reducing effect remain unclear. We examined the potential stress-reducing effect of music listening in everyday life using both subjective and objective indicators of stress. Fifty-five healthy university students were examined in an ambulatory assessment study, both during a regular term week (five days) and during an examination week (five days). Participants rated their current music-listening behavior and perceived stress levels four times per day, and a sub-sample ($n = 25$) additionally provided saliva samples for the later analysis of cortisol and alpha-amylase on two consecutive days during both weeks. Results revealed that mere music listening was effective in reducing subjective stress levels ($p = 0.010$). The most profound effects were found when 'relaxation' was stated as the reason for music listening, with subsequent decreases in subjective stress levels ($p \leq 0.001$) and lower cortisol concentrations ($p \leq 0.001$). Alpha-amylase varied as a function of the arousal of the selected music, with energizing music increasing and relaxing music decreasing alpha-amylase activity ($p = 0.025$). These findings suggest that music listening can be considered a means of stress reduction in daily life, especially if it is listened to for the reason of relaxation. Furthermore, these results shed light on the physiological mechanisms underlying the stress-reducing effect of music, with music listening differentially affecting the physiological stress systems.
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1. Introduction

Stress has the potential to be a major health threat facilitating the development of disease and illness (McEwen, 1998). As stress is a ubiquitous phenomenon in everyday life, it is necessary to develop interventions that can target stress in daily life and thus prevent its detrimental effects

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on health. It is assumed that music listening has health-beneficial effects, which might be mediated by its potential stress-reducing effect (Thoma and Nater, 2011). However, the empirical evidence on the stress-reducing effect of listening to music is far from being consistent, and the underlying psychobiological mechanisms of this potential stress-reducing effect — especially in daily life — remain unknown.

1.1. Music listening as a means of stress reduction in daily life

While the effects of music therapy (for an overview see Hanser, 2010) and music making (e.g., Bittman et al., 2013) on health-related outcomes have been studied in a plethora of (experimental) studies, the effects of mere music listening remain understudied. Particularly due to the fact that music is present in everyday life (Hargreaves and North, 1999; Juslin and Laukka, 2004; Krause et al., 2015), it is an important endeavor to study the effects of music listening in ecologically valid settings (Skanland, 2013). Results from research in experimental settings and in clinical contexts suggest that music listening has the potential to reduce stress, both subjectively and physiologically. For example, Sandstrom and Russo (2010) found that music high in valence and low in arousal positively affected recovery of heart rate and skin conductance levels after a stressor. Although most studies support the notion that music low in arousal is stress-reducing, Chanda and Levitin (2013) suggest that context factors, such as control over music selection (experimenter- vs. participant-selected), need to be considered as well. In contrast to experimental studies in which the context of listening to music is highly artificial, recent research efforts suggest that it might be more ecologically valid to study music in people's natural habitat (North et al., 2004; van Goethem and Sloboda, 2011).

Only a small number of studies have investigated music listening in daily life (Greasley and Lamont, 2011; Juslin et al., 2008; Krause et al., 2015; Randall et al., 2014; Skanland, 2013; van Goethem and Sloboda, 2011). These studies predominantly describe music-listening behavior, and to date, no study has directly investigated the effects of music listening on both subjective and physiological stress. Some studies do hint at the idea of a stress-reducing effect of music listening in daily life (Juslin et al., 2008; van Goethem and Sloboda, 2011). Juslin et al. (2008), for example showed that listening to music increased feelings of calmness. Still, research investigating this potential stress-reducing effect from a multi-dimensional perspective, exploring both subjective and physiological measures of stress, and considering both musical and nonmusical context factors in this relationship, is lacking.

1.2. Beyond self-reports: The effect of music listening on the physiological stress response

Music listening has been associated with a down-regulation of the hypothalamic—pituitary—adrenal (HPA) axis, as shown via reductions in the concentrations of cortisol in various experimental and clinical contexts (for an overview see: Kreutz et al., 2012), as well as with alterations of autonomic nervous system activity, with decreases in heart rate and blood pressure (for an overview see: Hodges, 2011). However, the pattern of results concerning biological effects of music is not consistent. Some studies failed to find an effect of music on stress-related physiological changes. Chlan et al. (2013), for example, found no effect of music listening on cortisol levels in mechanically ventilated patients. However, another study found a beneficial effect of music listening on parameters of the autonomic nervous systems in mechanically ventilated patients (Han et al., 2010). A plethora of studies (either investigating patients or healthy participants) report discrepant findings when comparing self-reports of stress to physiological markers of stress (DeMarco et al., 2011; Gerra et al., 1998; Thoma et al., 2013): for example, Thoma et al. (2013) found relaxing music to be less effective in reducing cortisol concentrations than the sound of rippling water, whereas there was no difference with regard to subjective stress levels. The same pattern emerged in the study by DeMarco et al. (2011) in which music reduced subjective stress levels but not heart rate and blood pressure in patients undergoing surgery. This heterogeneity of findings is probably due to methodological issues, such as different music selection and/or varying intensity of stressors, thus emphasizing the need to further study the effect of context variables on the stress-reducing effect of music listening.

1.3. The need to consider context variables: Valence/arousal, reasons for music listening, and stress intensity

To date, there is no empirically sound knowledge on which music (in terms of valence and arousal) is particularly effective for stress reduction purposes (Sandstrom and Russo, 2010). To the best of our knowledge, only one experimental study has systematically investigated the effect of the valence and arousal of music in the context of stress. In this aforementioned study, Sandstrom and Russo (2010) were able to show that recovery from a stressor worked best if participants listened to music which was low in arousal and positive in valence. However, in daily life, no study so far has related valence and arousal ratings to the stress-reducing effect of music listening

Juslin et al. (2008) suggest that the emotional effect of music varies as a function of reasons for music listening. Since 'relaxation' is one of the main reasons why individuals listen to music (Greasley and Lamont, 2011; Juslin et al., 2008; van Goethem and Sloboda, 2011), it seems plausible to examine reasons for music listening in the context of the stress-reducing effect of music listening. To date, no study has empirically examined whether reasons for music listening may affect the stress-reducing effect of music listening in daily life.

In her review on the stress-reducing effect of music listening, Pelletier (2004) summarized that music may only be effective in reducing stress in the context of a mild stressor as opposed to a strong stressor. This finding was corroborated by a recent experimental study from our group (Thoma et al., 2013), which found that listening to music prior to a (strong) socio-evaluative stressor was not effective in reducing stress, thus emphasizing the idea that the

stress-reducing effect of music may vary as a function of stressor intensity. However, so far, no study has examined the effect of real-life stressors of varying intensity on the stress-reducing effect of music listening in daily life.

1.4. Research question

Current evidence on music listening and stress is inconclusive. Many studies are set in experimental settings, thus limiting the generalizability of the results to daily life. However, as stress is a ubiquitous phenomenon and music listening is an activity of daily life, it seems of utmost importance to study the relationship between music listening and stress in an ecologically valid setting. As it remains unanswered which psychobiological mechanisms underlie the stress-reducing effect of music listening, it is necessary to combine subjective stress levels with physiological markers of stress to better understand under which circumstances what kind of music is effective in reducing stress.

Based on the available research findings, we hypothesized that music listening in daily life is associated with a subsequent reduction in stress, both subjectively and physiologically (Model 1). Furthermore, we were interested in whether context factors such as perceived valence and arousal of the selected music (Model 2), reasons for music listening (Model 3) and stress intensity (Model 4) influence the stress-reducing effect of music listening in daily life.

2. Method

2.1. Participants

A convenience sample with a total of 55 healthy students (35 female, 20 male) participated in the study. As females are over-represented in German Psychology classes, the proportion of females is higher as males in this study. Participants' ages ranged from 18 to 31 years with a mean of $M = 23.20$ ($SD = 3.11$) years. Four participants dropped out of the study after having completed the first measurement condition (due to technical problems or non-compliance). Recruitment was carried out by means of emails sent to university mailing lists. The eligibility criteria were as follows: German as native language, 18—35 years of age, BMI <30, fewer than five cigarettes per week, no current drug consumption, no intake of medication (except hormonal contraceptives), no chronic somatic or psychiatric disease, regular menstruation, no pregnancy, no breast-feeding. Participation in the study was voluntary and each participant received either 50 Euro or course credit as compensation. The study was approved by the local Ethics Committee of the Department of Psychology at the University of Marburg, Germany.

2.2. Procedure

The study used a within-subject design, in which participants were examined twice: at the beginning of a semester (henceforth referred to as control condition) and at the end of a semester, when exams are typically being taken in the German university system (henceforth referred to as exam condition) for five consecutive days each. During an introductory session, participants were familiarized with the handling of a pre-programmed iPod® touch (iDialogPad), on which they were required to complete six assessments on each day. The first assessment had to be triggered by the participant directly after awakening. Subsequently, a timer activated the next assessment after 30 min. Following this, assessments ensued at four time points: at 1000 h, 1400 h, 1800 h, and 2100 h. In case participants missed an assessment, they were reminded for up to four times every 5 min. The compliance rate was high in this study, with a relatively small number of missing data. The number of missing data was 1000 h assessment: $n = 159$ (i.e. 7.23% of all data points), 1400 h assessment: $n = 39$ (i.e. 1.77% of all data points), 1800 h assessment: $n = 37$ (i.e. 1.68% of all data points), 2100 h assessment: $n = 41$ (i.e. 1.86% of all data points).

A subsample ($n = 25$, 18 women) was asked to collect a saliva sample after each assessment on two consecutive days during both measurement conditions.

2.3. Measures

For the purpose of analyzing relationships between music listening and stress, only the assessments at 1000 h, 1400 h, 1800 h, 2100 h will be considered here, as only these assessments contained items on both stress and music-listening behavior.

2.3.1. Stress

Stress was assessed using a multi-dimensional approach: By means of real-time data collection, both subjective and physiological stress was measured.

Momentary subjective stress: Momentary subjective stress levels were measured using the item 'At this moment, I feel stressed', which was answered on a 5-point Likert scale ranging from 0 to 4. Low scores indicate low levels of stress and high scores indicate high levels of stress, respectively. In order to keep the compliance rate as high as possible we decided to use a single-item measure for stress which has recently been shown to have sufficient psychometric properties (Elo et al., 2003).

Physiological stress markers: Salivary cortisol (sCort) and salivary alpha-amylase (sAA) were measured. Cortisol is a hormone representing the activity of the hypothalamic—pituitary—adrenal (HPA) axis and is a valid biomarker of stress (Hellhammer et al., 2009). Alpha-amylase is an enzyme in saliva which can be considered as a biomarker for stress-related changes in the activity of the autonomous nervous system (ANS) (Nater and Rohleder, 2009). Both sCort and sAA can be concurrently measured non-invasively in daily life settings (e.g., Nater et al., 2013; Rohleder et al., 2008). In order to collect data on sCort and sAA, participants were asked to collect whole saliva using the SaliCap® system (IBL, Hamburg, Germany) by not swallowing for 2 min and then transferring all accumulated saliva into a pre-labeled vial via a straw. In order to control for participants' compliance, they were asked during each daily assessment on the iPod® to enter the number printed on the tube they were using. Participants were asked to store the tubes as cool as possible in their freezers or fridges at home. At the end of each assessment period, the tubes

were stored at −20 °C in freezers at the Biochemical Laboratory of the Department of Psychology, University of Marburg, until analysis. sCort levels were measured using a commercially available enzyme-linked immunoassay (IBL, Hamburg, Germany). sAA activity was measured using a kinetic colorimetric test and reagents obtained from Roche (Fa. Roche Diagnostics, Mannheim, Germany). Interassay variance was 10.9% for sAA and 6.5% for sCort, and intraassay variance was 6.5% for sAA and 6.2% for sCort.

2.3.2. Stress intensity

Furthermore, to assess the impact of stress intensity on the potential stress-reducing effect of music listening, participants completed the assessments during two conditions of varying stress intensity. Therefore, we chose to examine the stress-reducing effect once during a (less stressful) control condition at the beginning of a semester and once during a (more stressful) exam condition at the end of a semester. This decision was based on the evidence that academic examinations are a major contributor to students' stress levels (Abouserie, 1994).

2.3.3. Music-listening behavior

Ambulatory assessment: The first item on participants' music-listening behavior addressed the question of whether participants had been deliberately listening to music since the last assessment. Deliberate music listening was defined as 'actively selecting music and consciously listening to it'. For example, background music while shopping was not considered as deliberate music listening, whereas music listening via electronic devices was considered as deliberate music listening. Based on this item, the daily assessments were classified as 'non-music episodes' if no deliberate music listening was reported and 'music episodes' if deliberate music listening was reported. More music-related questions followed only if participants confirmed that they had deliberately listened to music. Participants were asked to evaluate the music on a visual analogue scale (0 to 100). The wording of the question was as follows: 'The music you have been listening to, was ...'. Participants indicated the perceived valence (ranging from 0 (sad) to 100 (happy)) and arousal (ranging from 0 (relaxing) to 100 (energizing)) of the music they had been listening to. Afterwards, they were asked to indicate reasons for listening to music. In accordance with the existing literature, we selected those reasons which have been most frequently reported in recent studies on music listening in daily life (Juslin et al., 2008; Linnemann et al., 2015; van Goethem and Sloboda, 2011): 'relaxation', 'activation', 'distraction', and 'reducing boredom'. For each reason, participants indicated whether it applied or not. Thus, multiple responses were possible.

Questionnaire: The Music Preference Questionnaire (MPQ) (Nater et al., 2005) was used to gain insight into participants' habitual music preference and situations in which they usually decide to listen to music. It comprises items measuring participants' music preference, their reasons for listening to music, and the average time they spend listening to music. Furthermore, on a scale ranging from 1 (not important) to 5 (very important), participants indicated the significance of music in their lives. This questionnaire was available online and participants were asked to fill it out during the first measurement period.

2.4. Data analysis

In order to account for the nested structure in the data, analyses are based on hierarchical linear modeling (HLM) (Raudenbush et al., 2004). Subjective stress, sCort, sAA, and items on music listening (music episode (yes/no), perceived valence, perceived arousal, reasons for music listening) were considered level-1 variables. In analyses concerning the relationship between music episode and sCort, time of day has been added as predictor ($UC = -0.45$, $t(363) = -16.57$, $p \leq 0.001$). However, in analyses concerning valence/arousal and reasons for music we refrained from adding time of day as predictor for two reasons: First, the number of level-1 observations is already relatively small, as only music episodes could enter analyses. Therefore we aimed at models which were parsimonious in order to avoid type-*B* error inflation. Second, as music listening was most often reported in the late afternoon and evening (both in the data set in which all observations were included ($UC = 0.03$, $t(2119) = 2.914$, $p = 0.004$) as well as in the data set in which only the observations were included from participants who provided saliva samples ($UC = 0.05$, $t(361) = 2.151$, $p = 0.032$)), we assume that the role of diurnal variations in sCort was kept to a minimum in these analyses. At the individual level (level-2), the intercept (β_0) was modeled as a function of gender (γ_{01}) and a residual component (u_0). In analyses concerning physiological parameters, body mass index (BMI) was entered at the individual level (level-2) due to its known association with HPA axis regulation.

Using HLM, both unconditional and conditional models were specified in accordance to the procedure described by Woltman et al. (2012). The comparison between these models was undertaken by means of χ^2-statistics, comparing the reduction in deviance as a measure of model fit. *p*-Values of ≤ 0.05 were considered significant. For all analyses, unstandardized coefficients (UC) are presented.

The subsample ($n = 25$, 18 female) that additionally provided saliva samples did not differ from the total sample either regarding music-listening behavior (number of music episodes: $M = 0.30$ vs. $M = 0.27$; $t(2090) = 1.51$, $p = 0.131$) or regarding stress levels ($M = 1.27$ vs. $M = 1.19$; $t(2090) = 1.72$, $p = 0.086$). Biological data were checked for normality using the Kolmogorov—Smirnov test. Both sCort and sAA were log-transformed due to non-normality using the formula $\ln(x) + 10$. In all tested models concerning sCort, women showed higher sCort concentrations than men ($UC \geq 0.29$, $t(22) \geq 2.19$, $p \leq 0.039$). BMI did not account for variation in sCort ($UC \geq -0.03$, $t(22) \leq -0.83$, $p \geq 0.413$). With regard to sAA, the reverse pattern emerged: Gender was not associated with sAA ($UC \geq -0.26$, $t(22) \leq -0.54$, $p \geq 0.595$), whereas a higher BMI was associated with higher sAA activity ($UC \geq 0.17$, $t(22) \leq 2.10$, $p \leq 0.047$). As $n = 13$ of the female participants took oral contraceptives, we tested whether the use of oral contraceptives influenced sCort concentrations and sAA activity. Neither sCort concentrations ($UC = 0.19$, $t(14) = 1.595$, $p = 0.133$) nor sAA activity ($UC = 0.18$, $t(14) = 0.297$, $p = 0.771$) were affected by the intake of oral contraceptives.

3. Results

Overall, music episodes accounted for 30% of the daily assessments. Music was listened to more often during the control condition (34%) than during the exam condition (25%), with condition explaining 1.09% of the variance in the frequency of music episodes ($\chi^2 = 33.23$, df = 3, $p \leq 0.001$). The reasons for listening to music were (in descending order): 'activation' (57%), 'relaxation' (47%), 'distraction' (47%), and 'reducing boredom' (46%) with 'relaxation' as reason for music listening being reported significantly more often during the late afternoon and evening than during the morning ($UC = 0.05$, $t(588) = 3.358$, $p \leq 0.001$). On average, the perceived valence of the music was rated as rather positive ($M = 68.89$, SD = 20.66) and the perceived arousal was rated as rather high ($M = 64.86$, SD = 24.97). Data from the MPQ revealed that participants listened to music on average for $M = 2.40$ (SD = 2.02) h per day, and participants rated music as important for their lives ($M = 4.21$, SD = 0.99).

3.1. Is music listening associated with lower stress ratings (Model 1)?

We first examined whether music listening was associated with reduced levels of subjective stress (Model 1a). The unconditional model included only subjective stress as dependent variable at level-1 and gender at level-2, and no music listening. We then specified the following conditional model: Subjective stress levels were modeled as a function of music episodes (yes/no) (β_{1j}) and a residual component (r_{ij}). As all level-1 observations were entered in these analyses, we modeled these analyses as random effects models. At the individual level (level-2), the intercept (β_0) was modeled as a function of gender (γ_{01}) and a residual component (u_0). In line with our expectations, participants displayed lower subjective stress after having listened to music (Table 1, Model 1a). The model fit (calculated as the reduction in deviance from the unconditional model to the model including music episode as a predictor) was significant, with 1.79% of the variance in subjective stress being explained by music episode ($\chi^2 = 15.51$; df = 3; $p = 0.002$). We then modeled sCort concentrations as a function of time point (β_{1j}), music episode (yes/no) (β_{2j}) and a residual component (r_{ij}) at level-1 (Model 1b). At the individual level (level-2), the intercept (β_0) was modeled as function of gender (γ_{01}), body mass index (BMI) (γ_{02}), and a residual component (u_0). There was no association between music episode and sCort concentration. Calculating an identical model with sAA activity as momentary outcome, no associations between music episode and sAA activity were found (Model 1c). Results are presented in Table 1.

3.2. Does the stress-reducing effect of music vary as a function of perceived valence/arousal of the selected music (Model 2)?

In a next step, we analyzed the perceived valence and arousal of the selected music with respect to stress ratings (Model 2). For this purpose, only time points when music was listened to (music episodes) were entered into the

Table 1 Hierarchical linear models predicting repeatedly assessed momentary stress (40 measures) (Model 1a), repeatedly measured salivary cortisol (16 measures) (Model 1b) as well as predicting repeatedly measured salivary alpha-amylase activity (16 measures) (Model 1c) by music episode (Model 1b) using restricted maximum likelihood.

	Model 1a: changes in subjective stress music episode		Model 1b: changes in sCort by music episode		Model 1c: changes in sAA by music episode	
Fixed effects	Unstandardized coefficients	SE (df), *p*	Unstandardized coefficients	SE (df), *p*	Unstandardized coefficients	SE (df), *p*
Intercept level-2	1.17	0.12 (48), *≤0.001*	11.95	0.13 (22), *≤0.001*	10.55	1.18 (22), *≤0.001*
Gender	0.20	0.14 (48), *0.153*	0.34	0.13 (22), *0.012*	−0.04	0.28 (22), 0.875
BMI[1]			−0.02	0.02 (22), *0.425*	0.16	0.16 (22), 0.011
Music episode (0/1)[2]	−0.17	0.06 (49), *0.010*	−0.01	0.01 (24), *0.915*	0.04	0.09 (24), 0.678
Time of day[3]			−0.45	0.04 (24), *≤0.001*		
Random effects	Variance component	SD (df), *p*	Variance component	SD (df), *p*	Variance component	SD (df), *p*
Intercept level-1	0.25	0.50 (47), *≤0.001*	0.05	0.23 (19), *0.012*	0.63	0.80 (19), *≤0.001*
Music episode slope	0.06	0.25 (48), *0.029*	0.02	0.13 (21), *>0.500*	0.00	0.06 (21), 0.447
Residual	0.86	0.93	0.36	0.60	0.55	0.74

[1] BMI was only entered in analyses predicting sCort and sAA.
[2] (0/1): 0 = no music episode, 1 = music episode.
[3] Time of day was only entered in analyses predicting sCort.

analyses. As the number of level-1 observations is already reduced in these analyses we did not model these analyses as random effects models as this would further reduce the number of level-1 and -2 observations. The following conditional model was specified: Subjective stress levels were modeled as a function of perceived valence (β_{1j}) and perceived arousal (β_{2j}) and a residual component (r_{ij}) (Model 2a). At the individual level (level-2), the intercept (β_0) was modeled as a function of gender (γ_{01}) and a residual component (u_0). Neither perceived valence ($UC = -0.00$, $t(526) = -1.378$, $p = 0.169$) nor perceived arousal ($UC = 0.00$, $t(526) = 0.782$, $p = 0.434$) was associated with subjective stress levels. The reduction in deviance of this overall model was not significant ($\chi^2 = 7.89$; df = 4; $p = 0.095$). Concerning sCort concentrations (Model 2b), neither perceived valence ($UC = 0.00$, $t(115) = 0.175$, $p = 0.862$) nor perceived arousal ($UC = 0.01$, $t(115) = 1.453$, $p = 0.149$) was associated with sCort concentrations. Concerning sAA activity (Model 2c), the perceived arousal of the music influenced sAA activity, as music which was rated as high in arousal predicted increased sAA activity ($UC = 0.01$, $t(110) = 2.272$, $p = 0.025$). The overall model explained 3.55% of the variance in sAA. The reduction in deviance was not significant ($\chi^2 = 6.38$; df = 5; $p = 0.270$). There was no association between sAA activity and the perceived valence of the selected music ($UC = -0.00$, $t(110) = -1.679$, $p = 0.096$).

3.3. Does the stress-reducing effect of music vary as a function of reasons for music listening (Model 3)?

In a third conditional model, we analyzed whether specific reasons for music listening accounted for any changes in subjective stress levels (Model 3a). Subjective stress levels were modeled as a function of 'relaxation' (β_{1j}), 'activation' (β_{2j}), 'distraction' (β_{3j}), 'reducing boredom' (β_{4j}), and a residual component (r_{ij}). Each specific reason for music listening was coded as 0/1, with 0 indicating that this reason was not chosen and 1 indicating that this reason was chosen, respectively. At the individual level (level-2), the intercept (β_0) was modeled as a function of gender (γ_{01}) and a residual component (u_0). Only music that was listened to for the reason 'relaxation' yielded lower subjective stress ratings ($UC = -0.45$, $t(524) = -5.430$, $p \leq 0.001$). In an exploratory analysis we examined whether music listening for the reason 'relaxation' only occurred when participants were experiencing particularly high or low levels of stress in advance. In this time-lagged analysis, we could show that there was no association between previous stress levels and music listening for the reason 'relaxation' ($UC = 0.05$, $t(426) = 1.827$, $p = 0.068$). Music that was listened to for the reason 'distraction' was associated with increased subjective stress levels ($UC = 0.35$, $t(524) = 3.652$, $p \leq 0.001$). The reasons 'activation' ($UC = -0.06$, $t(524) = -0.717$, $p = 0.474$) and 'reducing boredom' ($UC = -0.14$, $t(524) = -1.535$, $p = 0.125$) showed no associations with subjective stress levels. In this model, the predictor 'relaxation' explained 6.50% of the variance in stress levels. The reduction in deviance (from the model including 'activation', 'distraction' and 'reducing boredom' as predictors to the model additionally including 'relaxation' as predictor) was significant ($\chi^2 = 44.43$; df = 6; $p \leq 0.001$).

Concerning sCort concentrations (Model 3b), it was shown that in accordance with the results on self-reported stress, music that was listened to for the reason 'relaxation' yielded lower sCort concentrations ($UC = -0.48$, $t(113) = -3.513$, $p \leq 0.001$). In this model, the predictor 'relaxation' explained 12.44% of the variance in sCort, and the reduction in deviance (from the model including 'activation', 'distraction' and 'reducing boredom' as predictors to the model additionally including 'relaxation' as predictor) was significant ($\chi^2 = 26.54$, df = 7, $p \leq 0.001$). Concerning sAA (Model 3c), reasons for music listening were not associated with sAA activity ($UC < 0.28$, $t(108) < 1.789$, $p > 0.076$).

3.4. Stress intensity across the two conditions

There was a significant difference between the levels of perceived stress during the two conditions ($t(2050) = 14.46$, $p \leq 0.001$). Perceived stress levels were higher during the exam condition, with 9.29% of the variance in subjective stress levels being explained by condition ($\chi^2 = 204.47$, df = 3, $p < 0.001$). Cortisol concentrations differed between the two conditions ($t(363) = 2.13$, $p = 0.034$), with higher sCort concentrations during the exam condition, with 1.34% of the variance in sCort being explained by condition ($\chi^2 = 18.81$, df = 4, $p = 0.001$). There was no difference in sAA activity ($t(355) = 1.37$, $p = 0.172$) between the control and exam condition.

3.5. Does the stress-reducing effect of music vary as a function of stress intensity (Model 4)?

We analyzed the stress-reducing effect of music listening as a function of condition (Model 4). Therefore, the following conditional model was specified: Subjective stress levels were modeled as a function of condition (β_{1j}), music episode (yes/no) (β_{2j}), condition × music episode (β_{3j}), and a residual component (r_{ij}). At the individual level (level-2), the intercept (β_0) was modeled as a function of gender (γ_{01}) and a residual component (u_0). A marginally significant interaction of condition × music episode emerged ($UC = 0.17$, $t(1836) = 1.903$, $p = 0.057$). In this model, the predictors condition and condition × music episode accounted for 8.80% of the variance in stress in addition to music episodes alone. The reduction in deviance (from including only music episode in the model to including condition and condition × music episode in the model) was significant ($\chi^2 = 178.140$, df = 7, $p \leq 0.001$). Although the interaction term did not reach statistical significance with a pre-set alpha level of 0.05, we further investigated the nature of this interaction effects as McClelland and Judd (1993) state that detecting interactions in field studies is statistically more difficult than in experimental studies. Interaction terms were interpreted in accordance with Preacher et al. (2006), and music episode was considered the focal predictor, while condition was considered the moderator. Results are presented in Fig. 1. During the control condition, the relationship between music listening and subjective stress was significant ($\gamma_{1|0} = -0.1711$, $t(49) = -2.3862$, $p = 0.021$), whereas during the exam condition, the relationship between music listening and subjective stress was not significant ($\gamma_{1|1} = 0.002$, $t(49) = 0.0258$, $p = 0.980$).

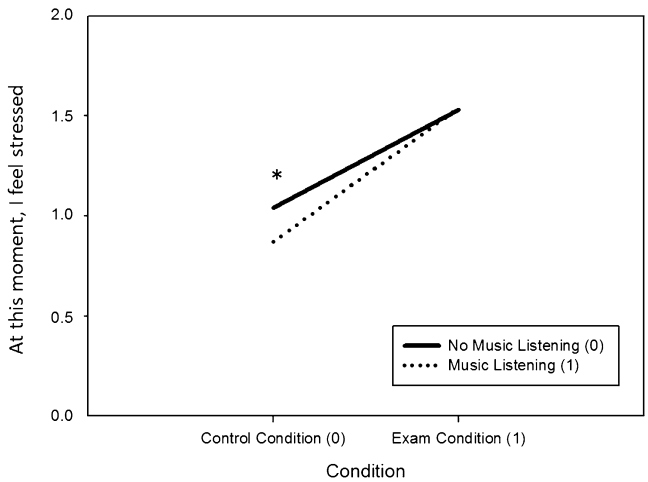

Figure 1 Mean stress level as a function of music listening (0/1) and condition (0/1). *Note*: * $p \leq 0.05$.

We then analyzed whether the effects of music listening on sCort varied as a function of condition. Concerning the number of music episodes, there was no significant interaction of condition × music episode ($UC = 0.01$, $t(329) = 0.050$, $p = 0.960$). With regard to sAA, there was no significant interaction of condition × music episode ($UC = 0.26$, $t(344) = 1.558$, $p = 0.120$).

4. Discussion

We sought to investigate the stress-reducing effect of music listening in daily life. Results suggest that mere music listening was associated with lower subjective stress levels. Nonmusical context factors contributed to the stress-reducing effect of music listening: Whereas the perceived valence of the selected music did not show any associations with stress parameters, music which was perceived as low in arousal was associated with lower sAA activity. The stress-reducing effect (attenuated subjective stress levels and concentrations of sCort) was particularly evident when music was listened to for the specific reason of 'relaxation' in the late afternoon and evening. There was an interaction between music and stress intensity insofar as music listening was only associated with lower subjective stress levels during the control condition.

4.1. Music listening reduces stress in daily life

Mere music listening had a profound effect on subjective stress levels, but it did not directly affect either HPA axis or ANS activity. At first glance, this appears to be in line with previous experimental studies, in which music listening reduced subjective stress but not its physiological correlates (e.g., DeMarco et al., 2011). However, our results provide evidence for the idea that different measures of stress may respond to different characteristics of music: sCort concentrations were lowest when the reason for music listening was 'relaxation', independent of its valence or arousal. The reverse pattern was true for sAA, which was lowest when participants listened to music that was perceived as low in arousal, independent of the reasons for music listening. Relating these findings to the existing literature, to the best of our knowledge, only one study has measured sCort and sAA simultaneously: In a laboratory study (Thoma et al., 2013), we showed that music listening prior to a stressor differentially affected HPA and ANS activity as well, with relaxing music accelerating ANS, but not HPA, recovery from a stressor. This is in accordance with our finding that the arousal of the music affected sAA but not sCort. Based on an overview of experimental studies, Chanda and Levitin (2013) suggest that relaxing music is usually associated with a decrease in cortisol. In our study, a reduction in cortisol was associated not with the *perception* of music as relaxing, but rather with the *reason* of 'relaxation'. Thus, our findings support the importance of a multi-system assessment of stress, as music listening differentially affects stress physiology.

Furthermore, our results shed light in the role of nonmusical context factors regarding the stress-reducing effect of music listening. This clearly extends notions from common models concerning the mechanisms of the effects of music on health. We recently proposed that the health-beneficial effects of music listening may be mediated by a reduction in stress (Thoma and Nater, 2011). Our results add to this hypothesis the importance of nonmusical context factors, especially reasons for music listening.

4.2. Relaxation as a reason for music listening is associated with the stress-reducing effect of music listening

It is well acknowledged that people listen to music in order to manage their emotions. Juslin et al. (2008) hypothesized that it is the reasons for listening to music which might explain the effects which music exerts. Our results provide empirical evidence that reasons for music listening, here 'relaxation', in conjunction with the time of day are relevant in the context of stress reduction. Although music listening for the reason 'relaxation' mainly occurred in the late afternoon and evening, when concentrations of sCort are generally lower, we could show that music listening for the reason 'relaxation' was associated with a reduction in stress independent of the previously stated stress intensity. However, as music listening was most often reported in the late afternoon and evening, it remains open to further investigation whether the time of day might be relevant to the stress-reducing effect of music listening. This is especially relevant as there is so far no sound knowledge on which timing of music listening relative to a stressor is best. Results vary depending on whether participants listen to music prior to or after a stressor (Khalfa et al., 2003; Thoma et al., 2013): concerning effects of music listening on HPA axis activity, listening to music after a stressor seemed to be more effective (Khalfa et al., 2003), whereas listening to music prior to a stressor positively affected ANS activity (Thoma et al., 2013). This somehow mirrors the findings in our study which show that music listening for the reason of relaxation, which mainly occurred in the evening, had a beneficial effect on subjective stress levels and HPA axis activity. Therefore, our results give rise to the assumption that time of day is relevant in designing music interventions

with probably most profound effects when music listening occurs in the late afternoon and evening.

Interestingly, in our study, listening to music for the reason 'distraction' was found to increase stress. This finding is worthy of attention, as it is discussed in the literature that music has beneficial effects by distracting people from aversive states (Bernatzky et al., 2001; Chanda and Levitin, 2013; Mitchell et al., 2007). However, such a beneficial effect has been predominantly examined in the context of acute stressors. Fancourt et al. (2014) suggest that effects of music may vary depending on whether people are experiencing acute versus chronic stress. Thus, one possible explanation for our finding is that music listening for the reason 'distraction' may be beneficial in the context of acute stressors but not under conditions of longer-lasting stress episodes, as reflected in our study by the exam condition.

At first glance, the finding concerning the limited stress-reducing effect under the experience of heightened stress seems to be in line with laboratory studies (for a review, see Pelletier, 2004). However, we were able to show that people listened to music less often in the exam condition, thus demonstrating that under heightened stress, music-listening behavior changes. It might be argued that participants did not find the time to listen to music while preparing for their exams. Therefore, we cannot conclude that music listening is not effective under the experience of heightened stress.

4.3. Limitations

Although this is the first study to investigate the effects of music listening on both subjective and physiological measures of stress in daily life, our findings need to be considered in the light of a number of limitations: We concurrently measured sAA and sCort at the same time, even though sAA and sCort underlie a different time course (Engert et al., 2011). However, in order to assess the direct effects of music listening on stress markers, it is necessary to collect sAA and sCort samples closer in time to when actual music listening occurred, for example by means of event-based sampling methods. Furthermore, only a subsample of participants provided saliva samples on two consecutive days during each condition. Therefore, it is possible that the proportion of variance in sCort and sAA through music listening might be overestimated. Although the subsample did not differ from the total sample in music-listening behavior and stress perception, these findings need to be corroborated in a larger sample. Furthermore, we assessed subjective stress with only one item. Although this is in accordance with Elo et al. (2003), it has to be kept in mind that the reliability of this stress measure might be lower than a more comprehensive assessment of subjective stress. With respect to the assessment of music-listening behavior, we only assessed *whether* music listening had occurred since the last assessment. Therefore, we do not have information on the duration and exact timing of the music listening. However, this information could be potentially relevant, especially due to the aforementioned specific reaction patterns of sAA and sCort to music listening. Furthermore, we did not collect data on the exact kind of music that was listened to with respect to genre, valence and arousal. However, 'objective' data on music listening might be necessary in order to further shed light on the health-beneficial effects of music listening in daily life. Concerning our study population, our inclusion criteria might have led to a selection bias as we constrained age to a range of 18 to 35 and further had an over-representation of females in our study. Also, our study population is quite homogenous when it comes to the educational background. All this limits the generalizability to the general population.

4.4. Implications for the use of music in both non-clinical and clinical contexts

Our results suggest that the stress-reducing effects of music listening are unfolded beyond the scope of experimental contexts with music listening differentially affecting stress physiology. The mechanisms behind the stress-reducing effect of music listening seem to vary between experimental contexts and daily life. In daily life, it seems to be the reasons for music listening that are associated with a stress-reducing effect of music listening. Particularly due to the fact that music is very popular, cost-effective and easily applicable in daily life, it seems promising to use music in daily life as a strategy for stress reduction.

Clinical populations seem to benefit from music therapy (Hanser, 2010). Whereas music therapy is a complex procedure involving a professional clinician, the mere listening to music in daily life might be a promising adjuvant treatment option for patients. As our study shows, it is not only music per se that has these stress-reducing effects; rather, a number of contextual factors also have to be considered in order to achieve maximum efficacy as an intervention. Interventions for the purpose of stress reduction should therefore concentrate on manipulating intentions behind music listening rather than music selection, and the timing of music listening relative to a stressor should be kept in mind.

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The funding sources had no role in the design of the study, data collection and analysis, or drafting of the manuscript.

Conflict of interest statement

The authors declare no financial interest related to the study.

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